

WASP-11b

R-1373

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_191122 · created 2026-08-01T01:04:53+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458809.7491 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.162 +/- 0.049
Transit depth (Rp/Rs)^2	2.62 +/- 1.6 [%]
Orbital Inclination (inc)	85.0 +/- 1.66
Airmass coefficient 1 (a1)	2.32 +/- 1.36
Airmass coefficient 2 (a2)	-0.56 +/- 0.3
Scatter in the residuals of the lightcurve fit is	62.13 %
Best Comparison Star	#2 - [198, 375]
Optimal Aperture	2.62
Optimal Annulus	12.99
Transit Duration (day)	0.043 +/- 0.039

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.162 ± 0.049 vs 0.131 (deviation 0.2σ)
Duration fit vs published	0.043 d vs 0.1075 d (deviation 0.53σ)
Mid-time O–C	6.3 min
Depth SNR	1.1
Residual scatter	62.13 %

Light curve

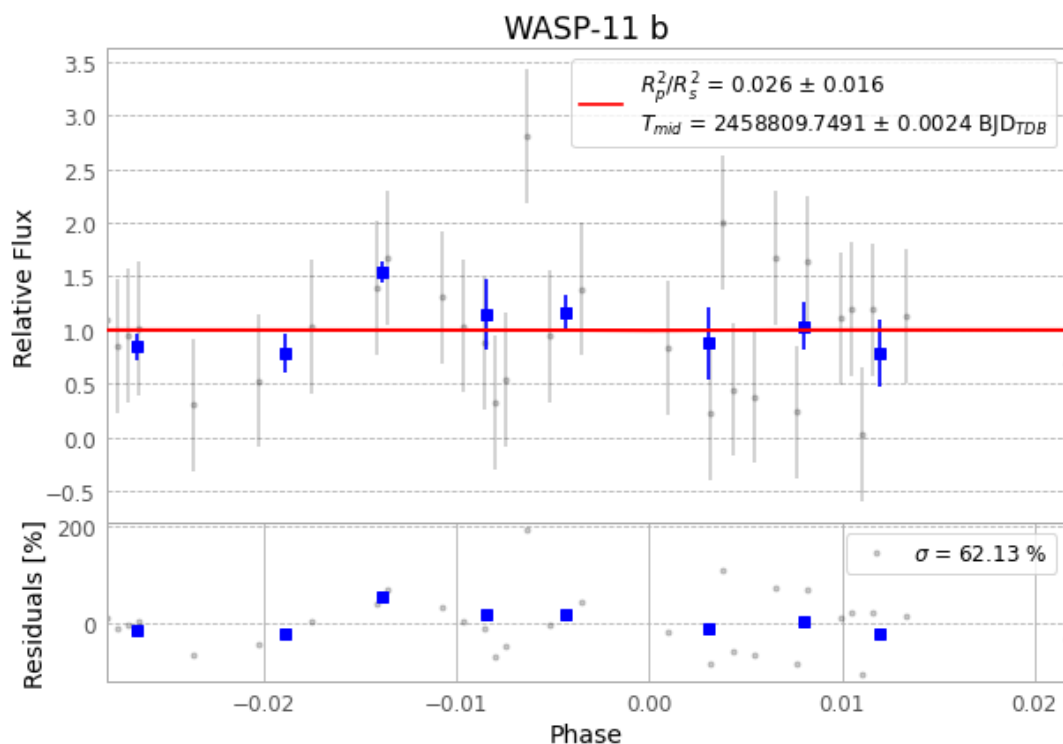


Figure 1 — FinalLightCurve_WASP-11 b_2019-11-21.png (SHA-256 09fa6465c5f75439...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [330, 432], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f11f7afe26773411...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

93 FITS frames (61 MB), HATP-10191122031812.FITS → HATP-10191122075412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-191122.log (9351ae94af27...), FinalLightCurve_WASP-11 b_2019-11-21.png (09fa6465c5f7...), FinalLightCurve_WASP-11 b_2019-11-21.csv (021d147b3091...)

Compute resources

Wall time 295.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 01:04Z.